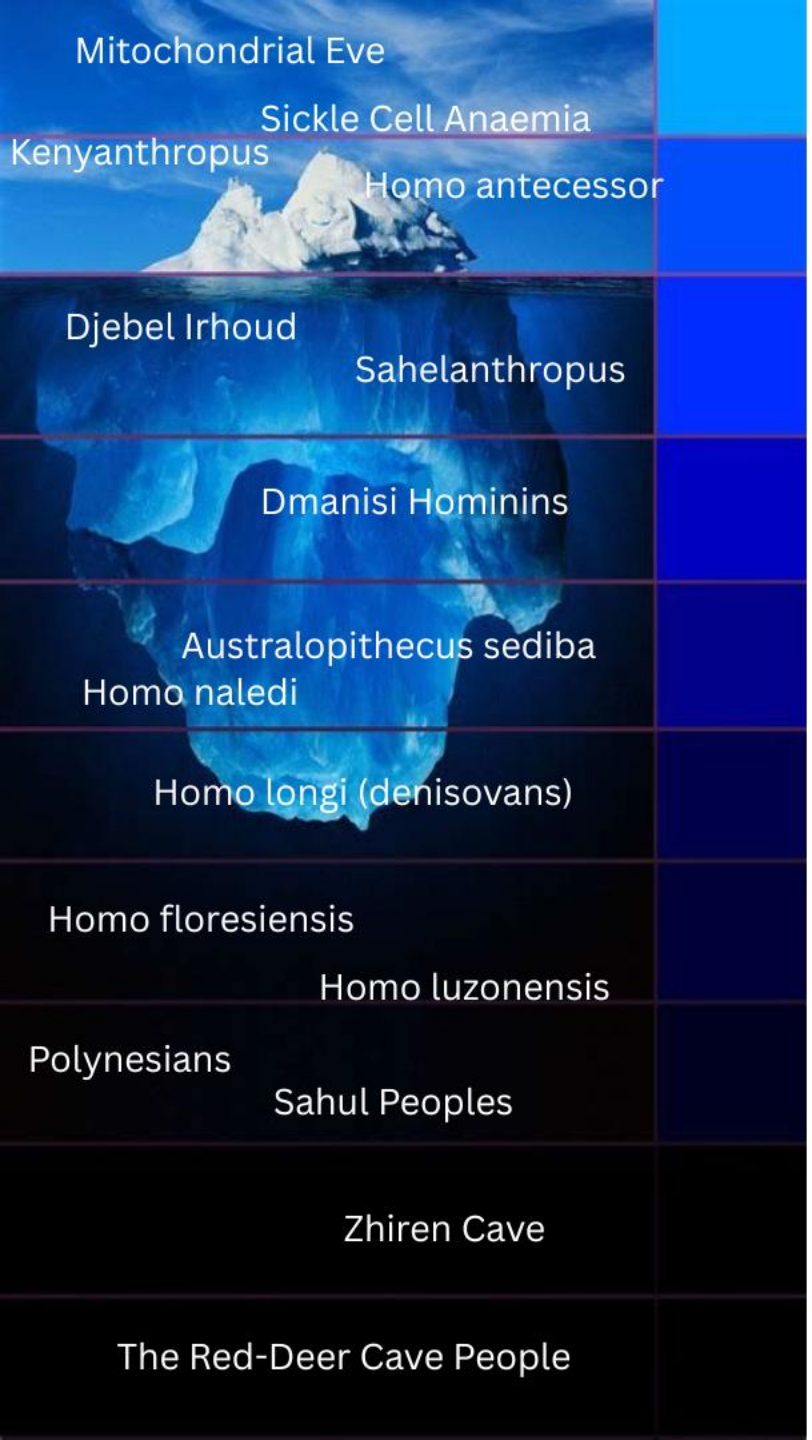


The Human Evolution Iceberg

- Anatomically Modern Human Migrations
- Genetics
- Introgression

(feel free to search some of the stuff on the left up, it's very weird)
(especially the last two)

Humans in Africa

- AMH don't truly appear until quite recently geologically speaking (~300kya)
- Djebel Irhoud, Morocco (~300kya)
- Omo Kibish, Ethiopia (~195kya)
- Herto, Ethiopia (~160kya)
- Ngaloba, Tanzania (~120kya)
- Most AMH features don't appear in a single linear step, they're more of a mosaic throughout the population of Africa.
- For example, many specimens have 1 or 2 major AMH features but are missing the rest.
- Regardless, early AMH are still closer to modern humans than archaic species such as neanderthals.

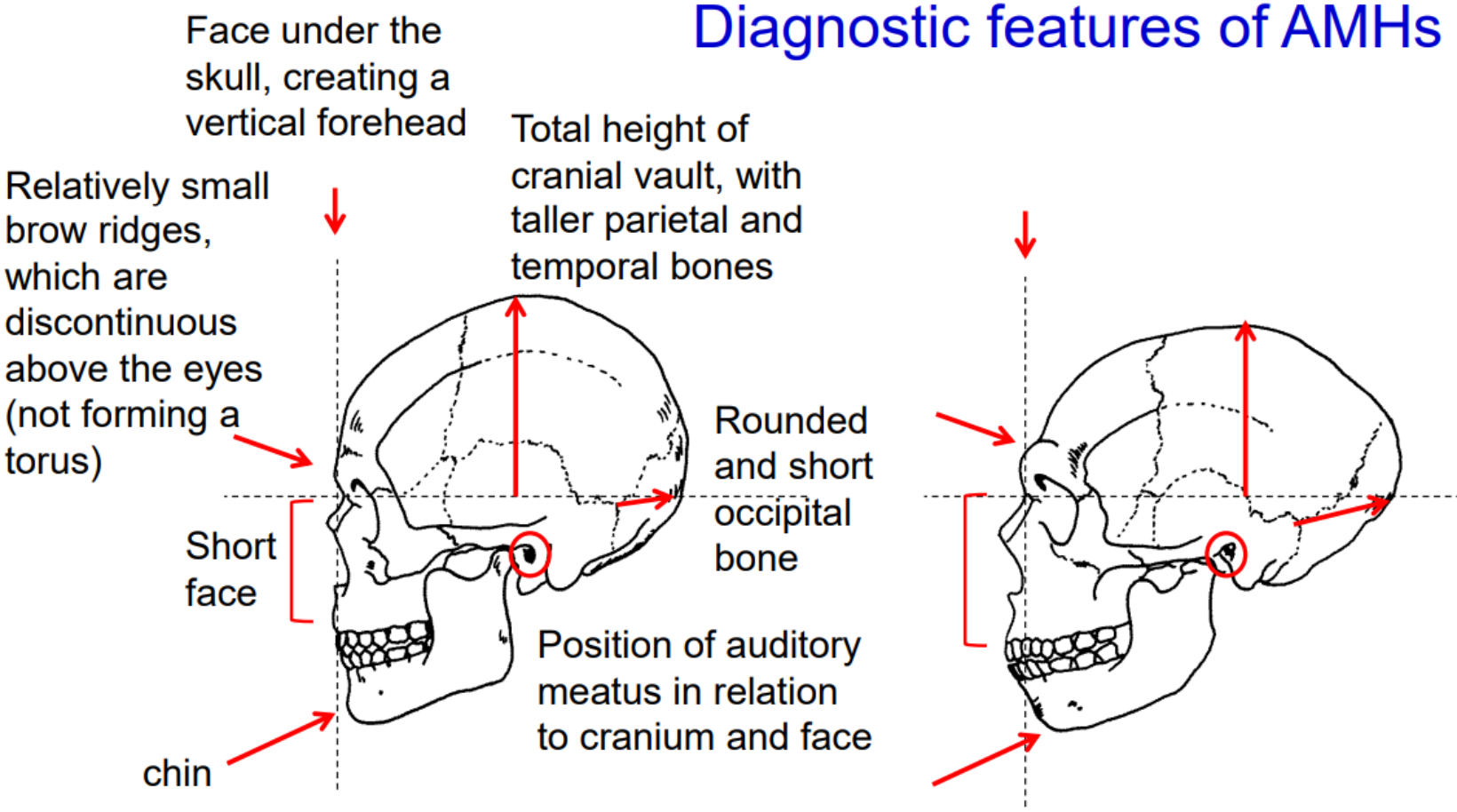
Djebel Irhoud

Omo Kibish I & II

Herto

Ngaloba

Diagnostic features of AMHs



Mitochondrial Eve/Beginning of AMHs

- Two hypotheses: Recent Africa Origin or Multi-regional evolution.
- Two types of data: Nuclear DNA (recombining) or Mitochondrial DNA (clonal)

Mitochondrial Eve – all living AMH descended from a single woman living between 100-200kya.

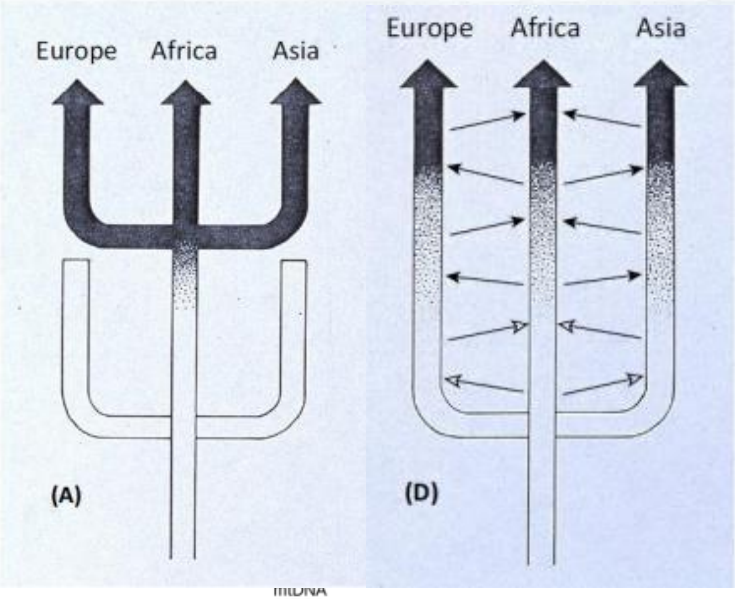
How?

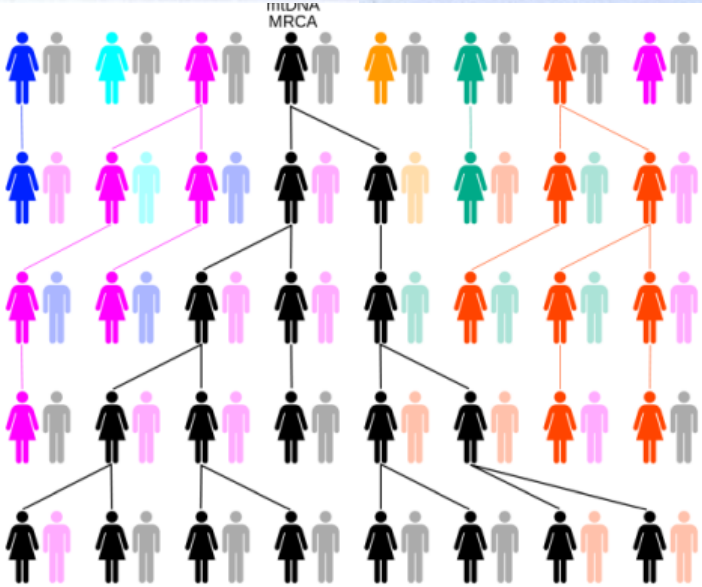
- Molecular clock analysis

- Assume constant mutation rate
- Calibrate with a fossil or aDNA (ancientDNA)
- Estimate hangs around 150-200kya

Same with Y chromosome?

- Not really, Y chromosome is hard to assemble but techniques are getting better





The structured population of Africa

Archaic anatomy ~13kya in Nigeria

- Suggests population remained highly structured in Africa until very recently.

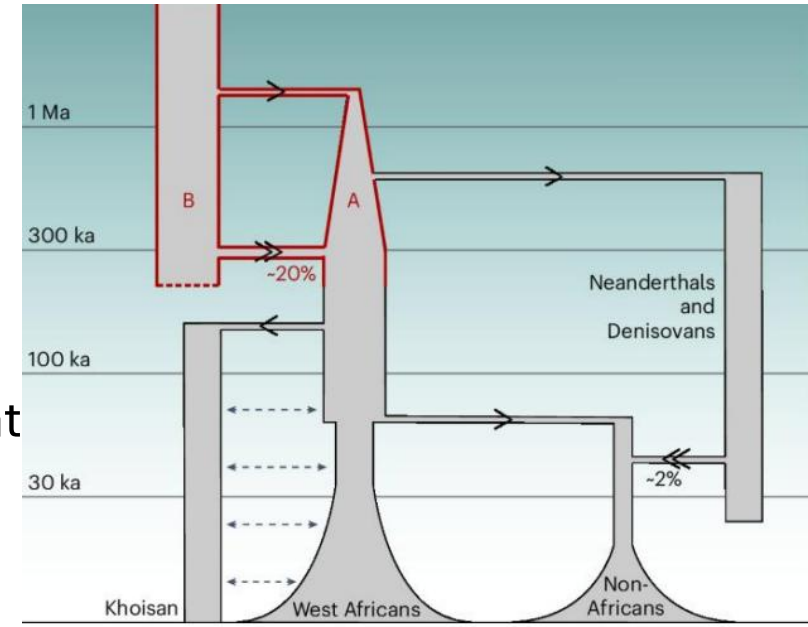
Divergent Y haplogroup (A00) only found in Cameroon (~239kya, 174-321kya).

- Suggests African populations have been deeply structured for a long time.

Looking at nuclear DNA:

- No consistent area of high diversity within Africa (this should be expected for a single origin point hypothesis)
- Instead, consistent with ancient population structure.
- Modern African population structure is essentially split into sub-Saharan and north African populations, where sub-Saharan populations are more diverse and contain less genetic admixture with Europeans and Asians.
- aDNA supports further deep population structure with no single origin.

Overall – African population likely began with extremely deep structuring that has been mixing continuously more and more until today. However, the structure was strong enough to still be evident even geologically recently (~13kya).



Out of Africa/African Origin

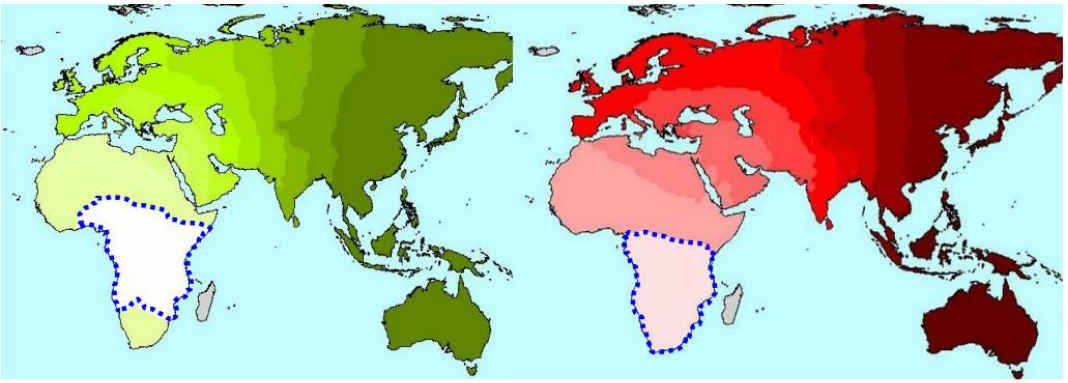
A – RAO (Recent African Origin)

B – RAO with hybridisation

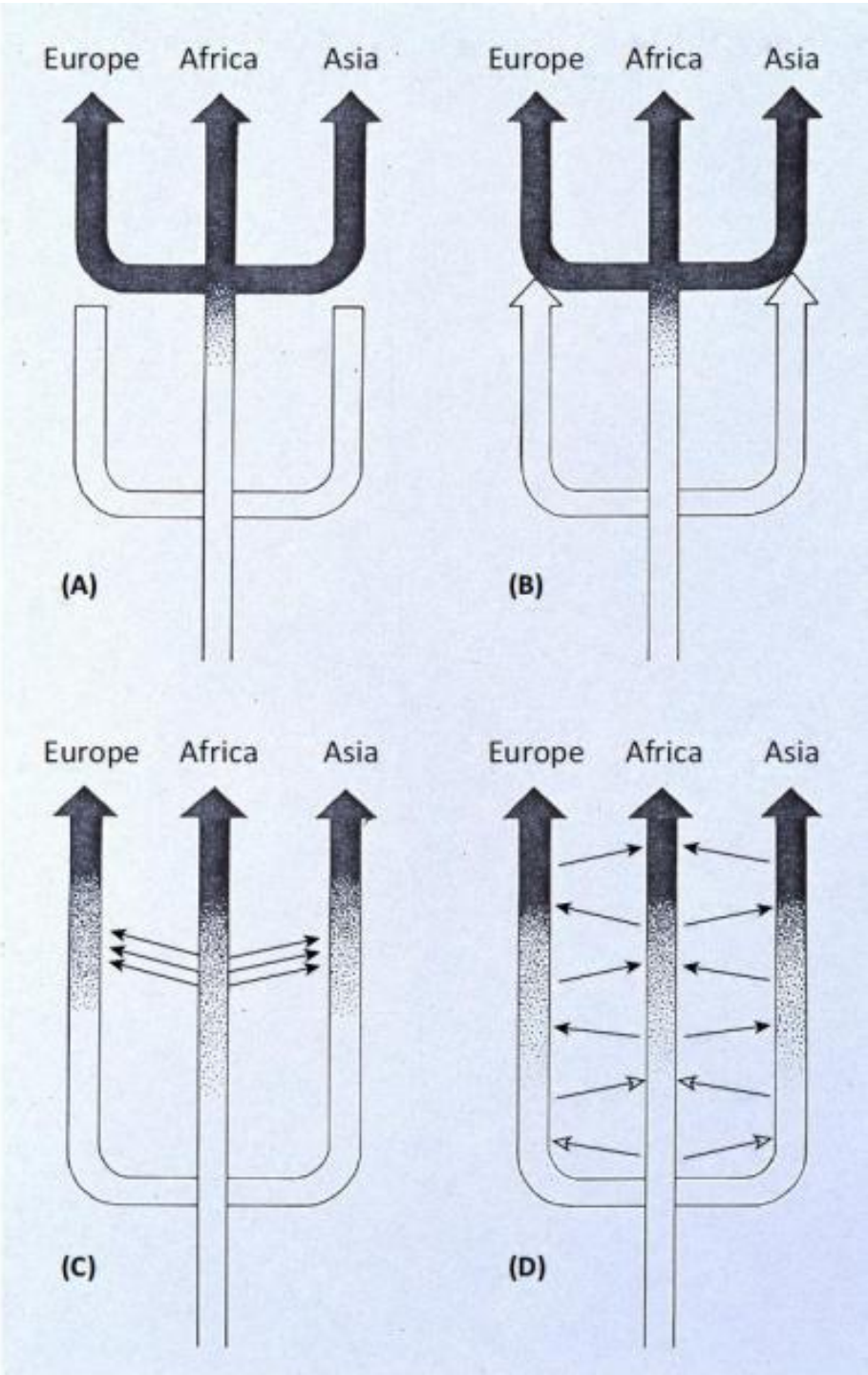
C – assimilation model

D – Multiregional evolution

- Genetic diversity decreases with distance from sub-Saharan Africa
- Both genes and morphology agree the origin of expansion out of Africa was sub-Saharan



Microsatellites Skull morphometrics

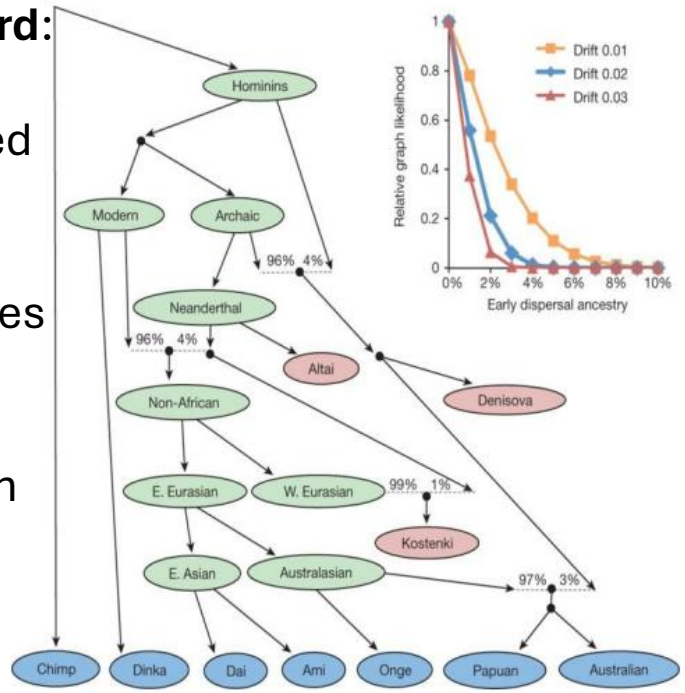


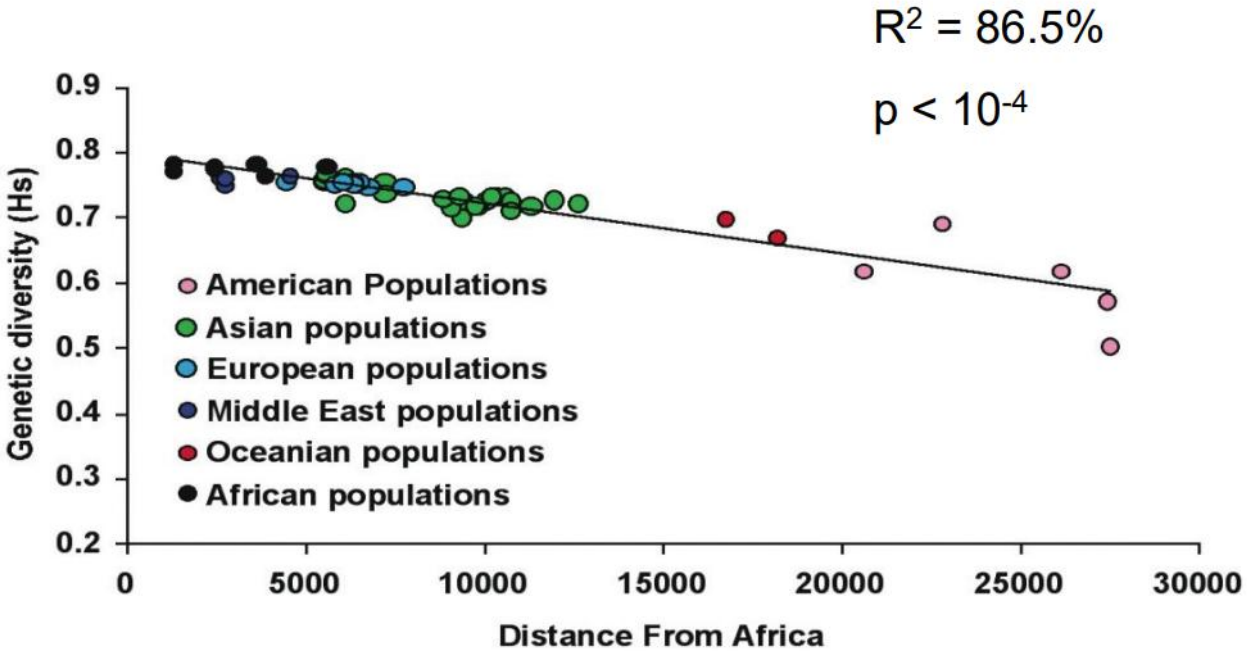
Single or Multiple exits?

- Possible exit times (low sea level/favourable climate): 130Kya, 60-70Kya
- Two routes, northern and southern
- **Northern** favourable between **130-90**Kya, **80-90**Kya
- **Southern** favourable between **130**Kya, **60-70**Kya
- **MitDNA** is consistent with ROA/single exit.
- **Whole genomes** suggest single origin, but a few % of early dispersals can go unnoticed – also NO DATES using admixture graph model.
- Sahul Peoples (Auzzie Aboriginals & Highland Papuans) possible descendents of first wave despite no model support.
- **HOWEVER**, recent paper finds support of an earlier exit for them after sequencing eDNA (Francesca Gandini et al. 2025 DOI: 10.1126/sciadv.ady9493).
- Use of haplotypes instead shows Papuans have an older component – consistent with earlier dispersal – BUT this has shown to be possibly down to phasing errors.

Niche Changes/Fossil Record:

- Australian Aboriginals appear very early compared to other groups so far from Africa.
- Niche expansion is the types of habitat suitable for hunter-gatherer humans expanding (grasslands with specific megafauna)
- **Major niche expansion ~70Kya**, just before major dispersal





Genetic Evolution in human lineage

Comparing whole genomes:

- Chimps and Humans have about ~1.23% SNPs divergence & ~3% ins/del divergence – total divergence of ~4%
- Problems occur with intraspecific variation/divergence & polarity/where in the lineage the divergences occurred.
- Neither of these can be understood with just two species' genomes.
- Secondly, even if you include the rest of the ape lineage, they are all very specialised and diverse so how do you know which one contains the ancestral sequences?
- Even if Gorillas and Chimps have the same, seemingly "ancestral" SNPs or mutations – how do you know that didn't arise from convergence and Homo has the ancestral gene?

Brain Size

- ASPM, CDK5RAP2, & CENPJ can affect neurogenic mitosis.
- MCHAP1 (DNA damage response pathway) can affect brain development.
- Mutations in these can cause microcephaly.
- dN/dS ratio can detect selection (S=silent/neutral mutations, N=non-silent mutations)
- The ratio between the two suggests selection if you assume all deleterious non-silent mutations were selected out of the genome.
- dN/dS ~ 1 = neutral selection – no genes under selection, mutations can occur anywhere
- dN/dS > 1 = natural selection – promotion of amino acid change
- dN/dS < 1 = purifying selection – preventing amino acid change

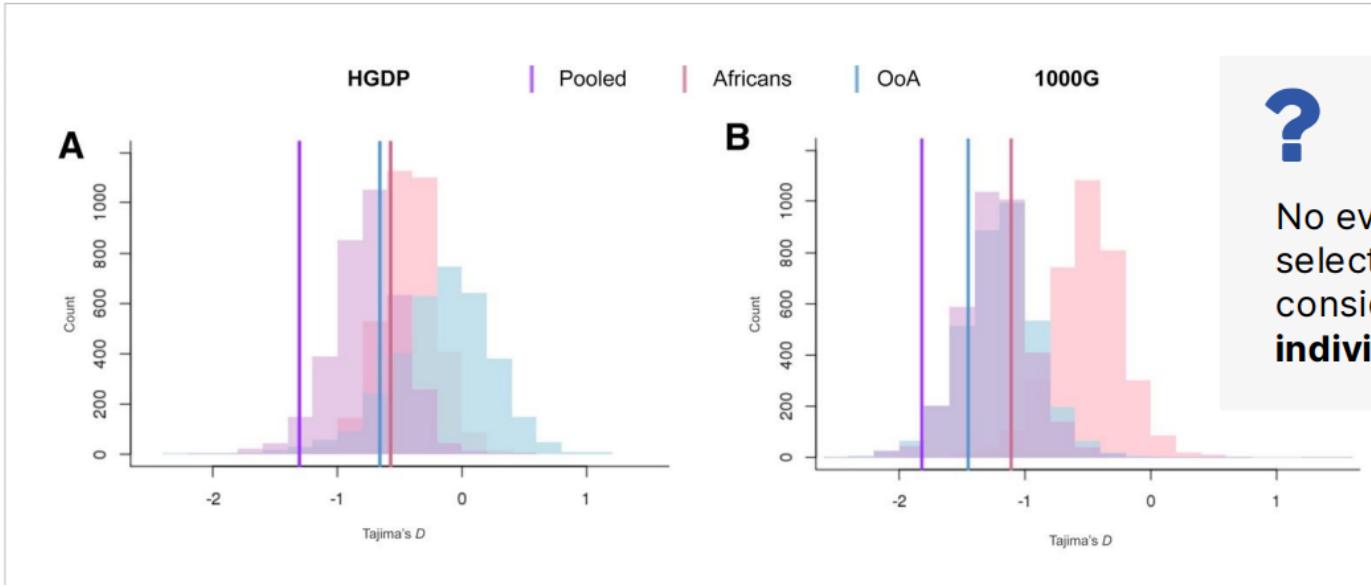
a)	Model 2			Model 0		
Gene	d*N*/d*S* in lineages leading to *Homo*	d*N*/d*S* in non-Human lineages	lnL(M2)	lnL(M0)	Likelihood Ratio Statistic	*P* value
ASPM	0.489	0.466	−21351.430	−21351.451	0.042	0.838
CDK5RAP2	0.729	0.676	−8718.412	−8718.440	0.054	0.816
CENPJ	0.623	0.503	−8664.800	−8664.986	0.373	0.542
MCPH1	0.793	0.579	−7301.153	−7301.505	0.704	0.401

- Analysis showed that there was no significant difference between selection for brain size changes in the lineage leading to homo specifically
- **All Anthropoids experienced brain-size selection**

What does this mean for brain size for example?
If genes associated with brain size in humans were under natural selection while in chimps, they were under purifying/neutral it would suggest positive selection for increased brain size in humans/chimps hold ancestral genes.
If the reverse was true and chimps were under natural selection while humans were under purifying or neutral it would suggest humans hold the ancestral sequence.
Obviously, with brain size the former is more likely true, but that's not necessarily the case for everything, for example – bipedality vs knucklewalking.

Language

FOXP2 – gene for language



No evidence of recent selection when considering **African individuals** - why?

- Shows two major amino acid changes in lineage leading to Homo
- Tracked to 200-100Kya – coinciding with emergence of AMH

HOWEVER

- No selective sweep in Africans
- Denisovans & Neanderthals show same changes – change likely predated 200-100Kya
- D statistic (Atkinson et al. 2018) showed selective sweep signal was caused by demographic bottlenecks. When included, evidence for recent selective sweep disappears.

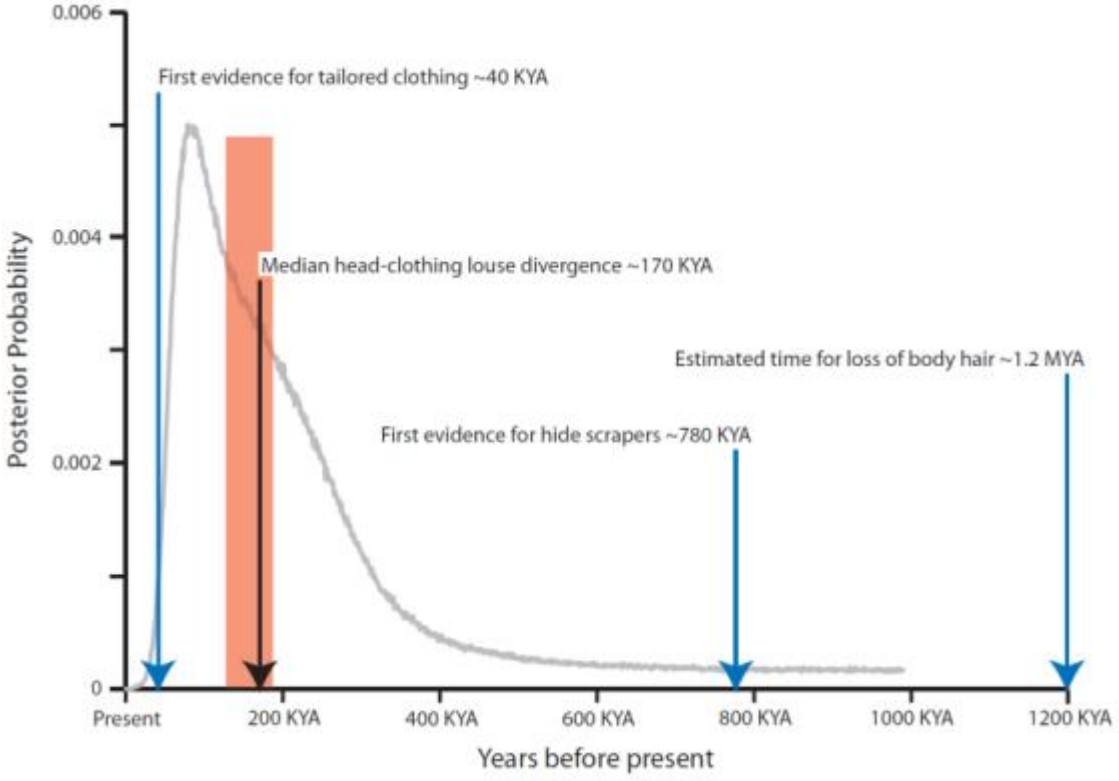
Clothes & Hair

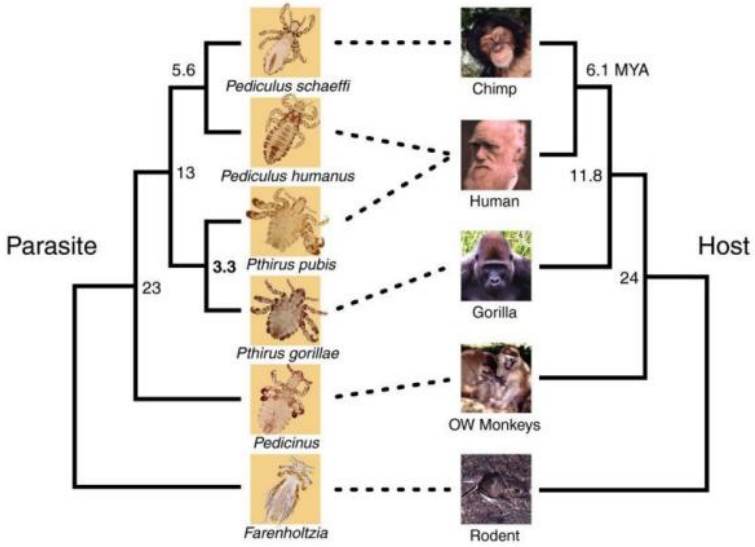
Hairlessness

- MC1R (involved in melanosomes/suggests skin exposure) diverged with chimps ~1.2Mya
- Split between head & pubic lice ~ 3.3Mya

Clothing

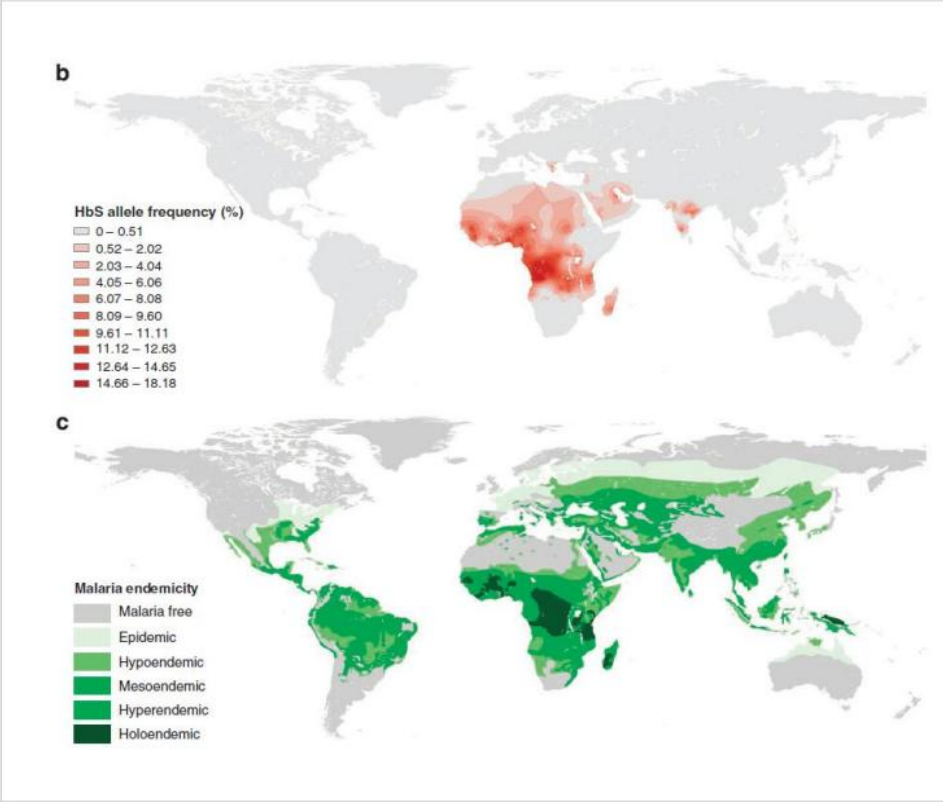
- Head and clothing lice split 170Kya
- Hide scrapers 780Kya
- Likely lots of contact with other hominin groups muddied the water/divergence times
- Dmanisi hominins (early H. erectus ~1.9Mya) – suggest clothing in these early groups as how else do they survive in a colder climate so early on? Maybe fire, but that's another can of worms.





Continued Human Evolution?

- African sickle-cell anaemia – heterozygote immunity to malaria
- Lactose tolerance



Methods:

- Selection scans
- GWAS

Demography & selection

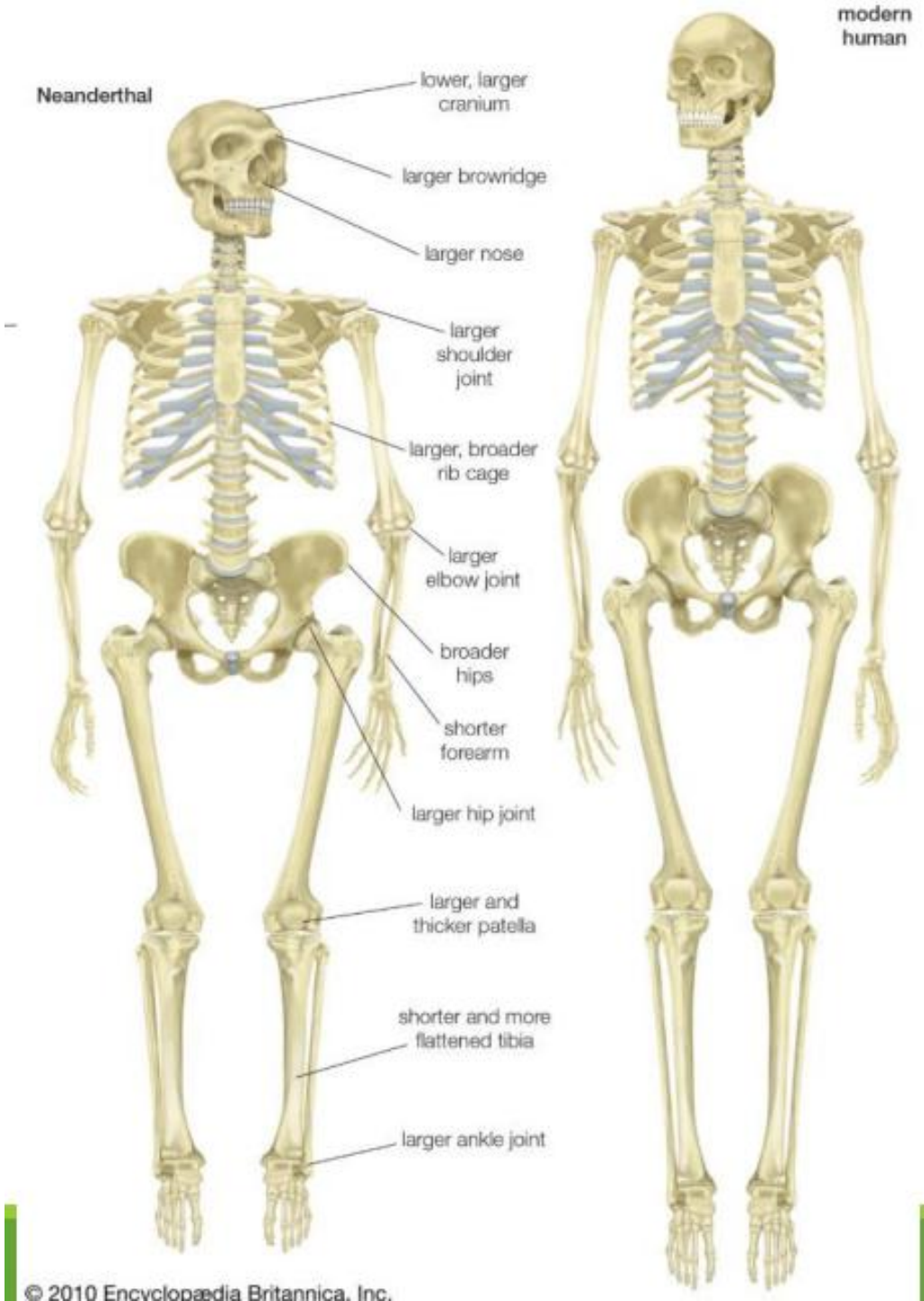
- Demographic distributions and bottlenecks often mimic selective sweep signals making it hard to make robust conclusions from the above methods.
- aDNA can sometimes unblur things as it fills in missing parts on the stems of trees and can distinguish between selection and founder effects

Conclusion – selection is likely still continuing but on differential reproductive success rather than survival – for example dutch people are probably tall because of sexual selection for those traits.

- In general, it's hard to detect anything concrete overall, but subpopulations of humans probably do still undergo selection even if it's not corelated with survival directly.

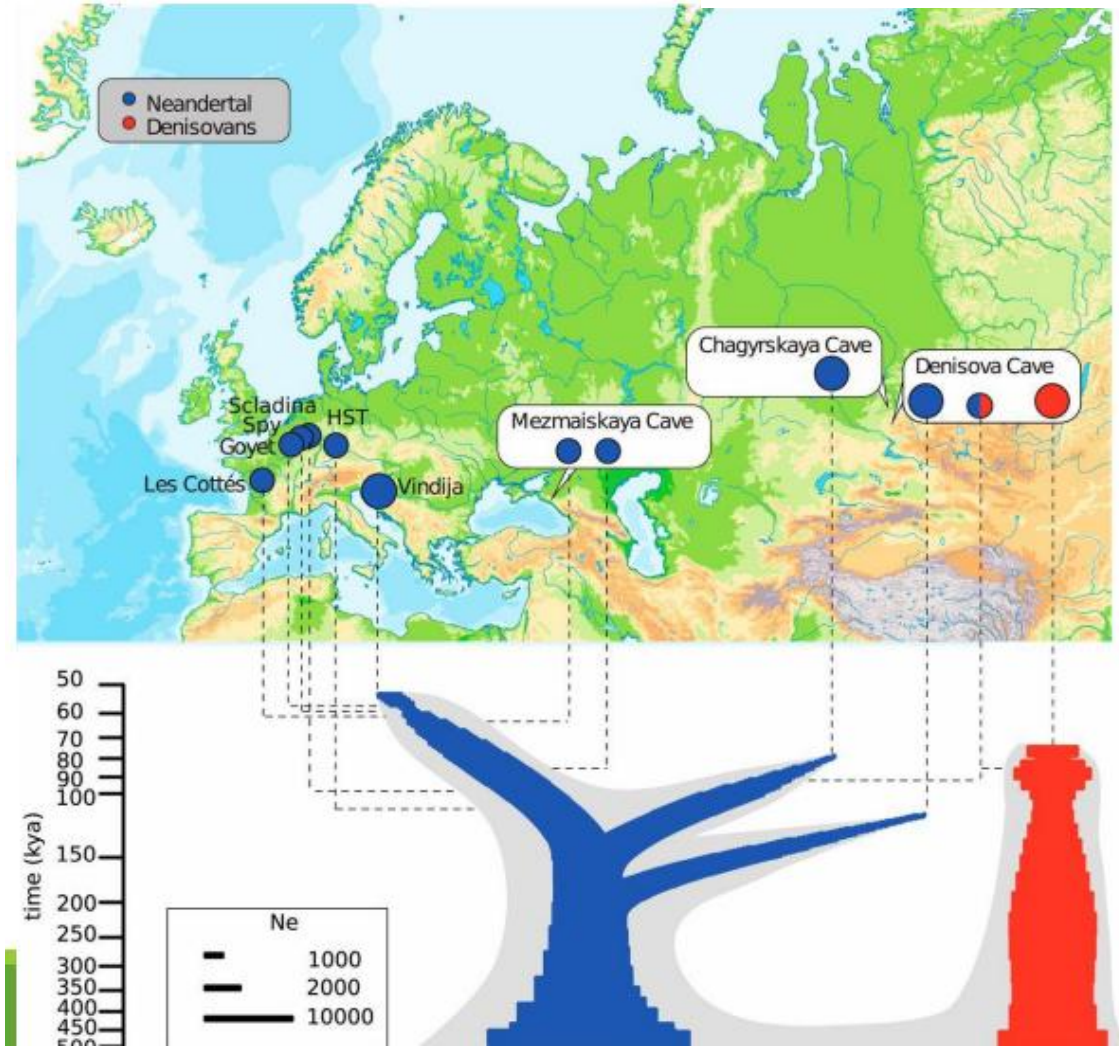
Neanderthals

- Probably split with us ~800Kya ("heidelbergensis" – BIG can of worms)
- Are considered to have appeared ~400Kya (protoneanderthals ~600Kya)
- Stocky robust build for colder climates
- Smaller frontal lobe, enlarged occipital lobe
- Small isolated groups
- Very successful until decline ~100Kya
- Advanced stone tool culture (Mousterian & Chatelperronian - same as early homo-sapiens)



Denisovans (Homo longi?)

- Known from only DNA and tiny fragments for ages
- More closely related to Neanderthals than Sapiens (split ~450-500Kya)
- Facial recontruction from methylation of eDNA found in Denisova cave
- The Harbin Cranium (Homo longi) appears to match the Denisovan dna quite well.

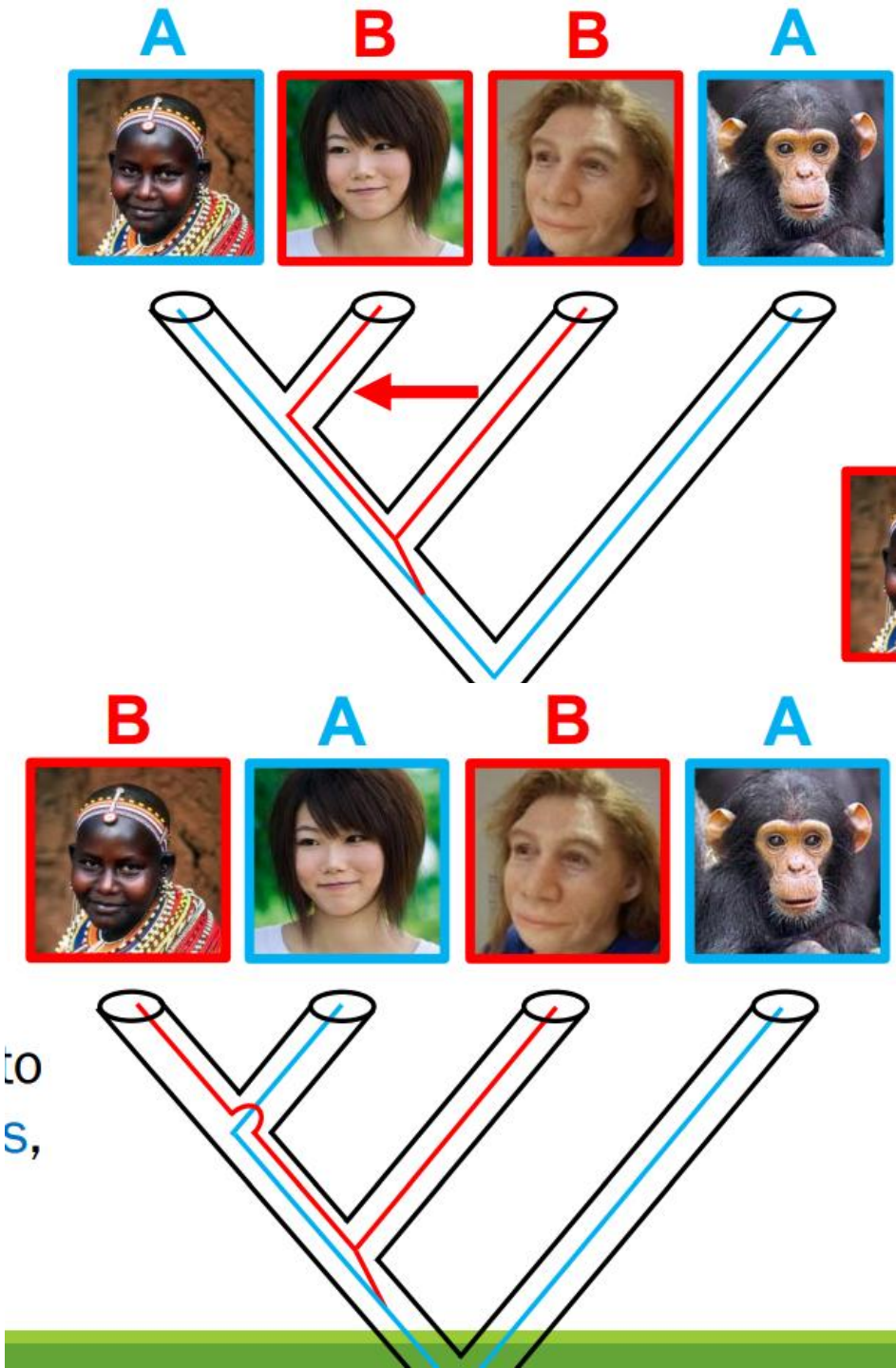


Introgression – Uh Oh

- ABBA-BABA (D-statistic) as a measure of introgression.
- Positive signal (1-4%) in Eurasian populations between neanderthals and sapiens
- Another positive signal with Melanesian populations and denisovans

HOWEVER

- No surviving neanderthal mitochondrial sequence or Y chromosome
- ABBA-BABA relies on constant mutation rate – which is not constant across human populations
- When taken into account, suggests signal is an artefact of bottlenecking (again) – see Bill Amos' stuff https://www.zoo.cam.ac.uk/directory/william-amos



Introgression at all? & adaptive introgression

- There was DEFINITELY hybridisation – we find hybrid individuals of sapiens, denisovans, and neanderthals in the fossil record – **BUT DID THEY CONTRIBUTE?**
- Did it have much of an effect? – Maybe? But in my opinion, it's probably not a huge effect, maybe ~0.1% (I have had too many conversations with Bill Amos though).

Adaptive?

- 40% Tibetans have Denisovan EPAS1 – beneficial for haemoglobin at high altitude – Bill Amos however lowk disproves this.... 🤨
- Neanderthals likely were the original carriers of genes associated with red hair
- Introgression signals appear to be lower in certain regions of the genome – suggesting negative selection (FOXP2, X chromosome hybrid incompatibilities, selection on males)
- Neanderthal introgression appears to be strongly sex-biased with most introgression stemming from Male neanderthals...
- (why no y chromosome then though?)

CARNIVORA!!!!!!!!

- **Stem carnivorans** close to the crown were **arboreal** - same with the early stems of caniforms and feliforms AND stem felids, ursids, and canids themselves.
- Essentially the default for modern carnivorans was usually arboreal and cursoriality evolved several times.
- **Today they occupy A LOT of niches**

- Bears, Dogs, & most Cats = Cursorial
- Some felids, viverrids (civets), linsangs = Arboreal
- Mustelids (weasels) & Herpistidae (mongoose) = Fossorial
- Some viverrids, felids, mustelids, herpistids, & ursids = Scansorial
- Pinnipeds (seals and sea lions) = Aquatic.

- **Occupy a lot of different diets too**

- Frugivorous, Carnivorous, Omnivorous, Myrmecophagous, Piscivorous, Herbivorous

Carnivora

THIS IS IN THE BROAD SENSE/RECENT ANCESTRALLY

Stealth, Power, Climbing

Cursoriality

"Creodonta"
(Polyphyletic)

"Miacidae"

Euungulata/Pholidota
Oxyaenidae
Hyaenodontidae
Viverravidae
Miacis
Nimravidae
Nandinia
Barbourofelidae
Machairodontidae
Felinae
Pantherinae
Viveridae
Hyaenidae
Eupleridae
Herpistidae
Amphicyonidae
Canidae
Ursidae
Pinnipedia
Alluridae
Mephitidae
Procyonidae
Mustelidae

Felidae
Feloidea

Viverroidea

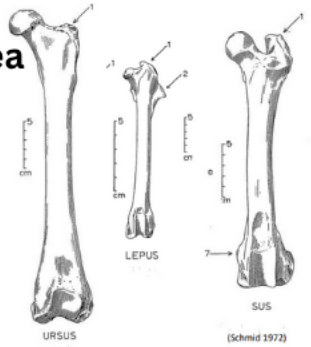
Musteloidea

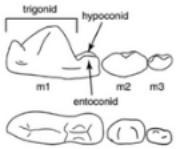
Carnassial apparatus
- Upper 4th premolar (P4)
- Lower first molar (M1)

Feliformia

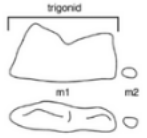
highly developed turbinates
anterior facing orbits

Caniformia





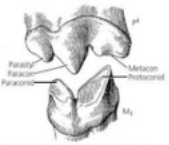
Generalist carnivoran



Holliday & Steppan 2004

Hypercarnivores
- Trigonid (m1) enlarged
- Absence of talonid
- Parastyle (P4) enlarged

→ Increase of shearing edges, decrease in grinding surfaces



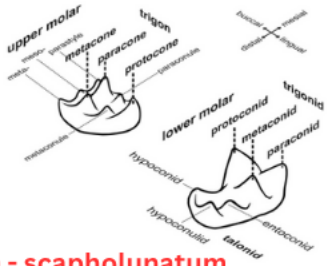
Westerheide & Rieger 2010

Carnivora

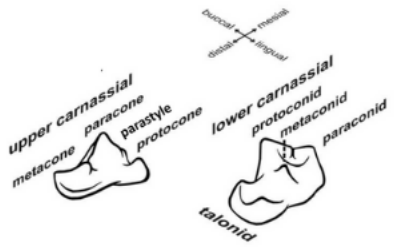
fused scaphoid & lunate - scapholunatum
reduced collar bones
lack 1st trocanter on femur (sometimes)
ossified auditory bulla (ecto+entotympanic)

Carnivoramorpha

Tribosphenic molar crowns



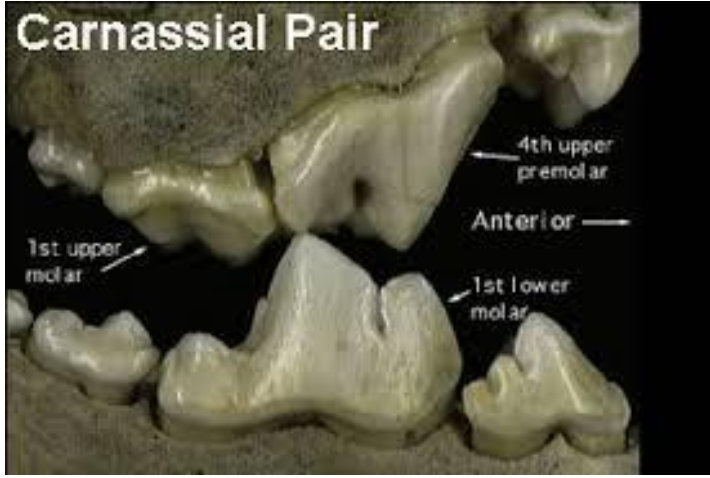
Carnassials of Carnivoramorpha



P4: enlargement of metacone/metastyle crest & reduction of protocone

m1: enlarged ridges of protoconid & paraconid (trigonid = shearing blade)

Anatomy

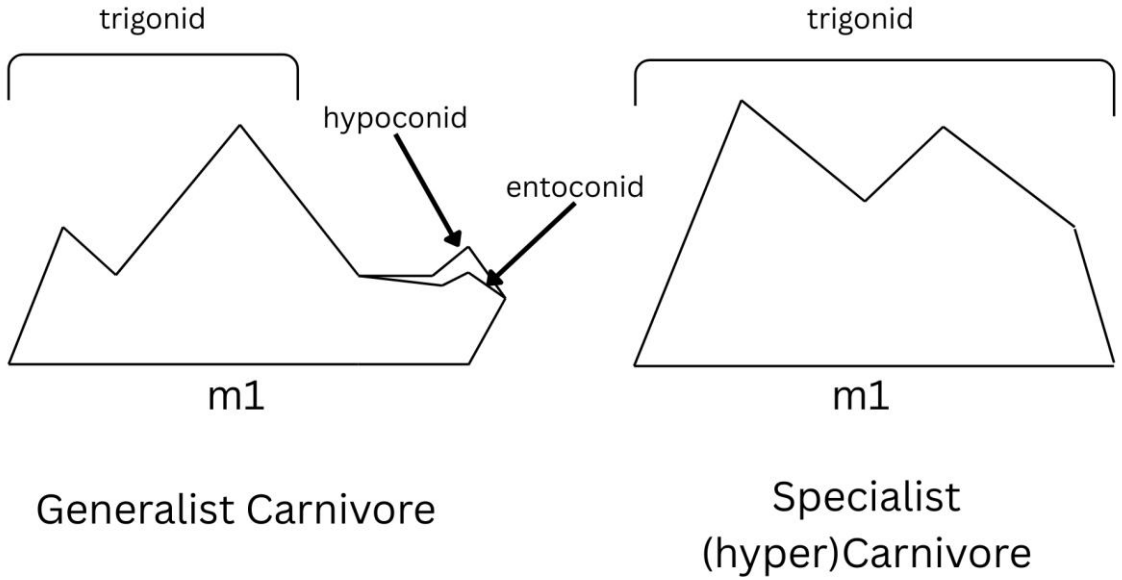


Carnassial apparatus – P4 & m1 (deciduous precursors too)

- P4 – enlarged metacone-metastyle crest + reduced protocone
- m1 – enlarged ridges of protoconid & paraconid (trigonid = shearing blade)
- In hypercarnivores grinding surfaces are decreased (talonid absent) and shearing surfaces increased (trigonid & parastyle enlarged)
- Canids generally retain the ancestral generalist conditions
- Felids generally enlarge their carnassials and reduce their molars – often resulting in the hypercarnivore anatomy
- This leaves felids in an evolutionary trap where they cannot return to generalist lifestyles – **MACHAIRODONTIDS**
- In bears many species nearly lose their carnassials as they have a diet full of vegetation – cave bears & pandas especially

Other anatomy

- Scapholunatum – fused scaphoid and lunate in carpus
- Reduced clavicle
- Lack of 3rd trochanter on femur
- Ossified ento/ectotympanic bulla
- Anterior pointing eyes (binocular vision)
- Highly developed turbinates

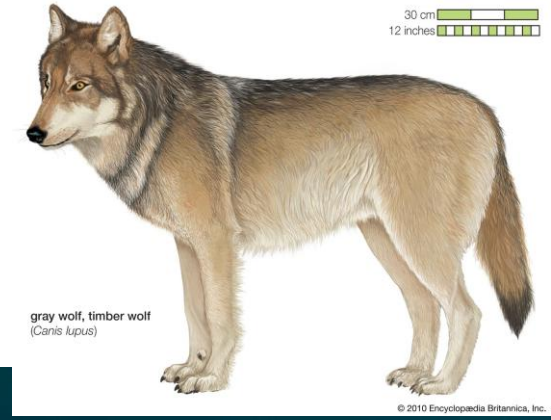


Carnivore adaptations

- Jaw joint & cheek teeth on same level to act like scissors
- Cylindrical mandibular condyle, restricts movement & disarticulation
- Large temp. fossa, temporalis, coronoid process, sagittal crest – strong bite force & stability of joint
- Small zygomatic fossa & angular process (reduced masseter)

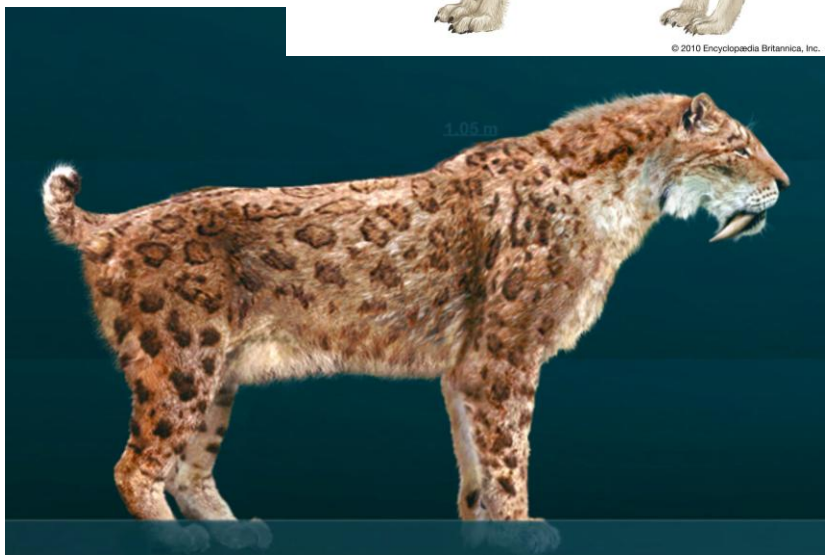
Felids – tend to be more arboreal ambush predators that use power and strength to take down prey

- Suppination of paws, retractable claws.

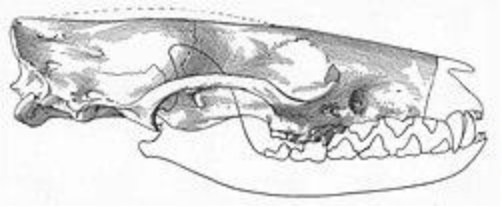


Canids – tend to be more cursorial, run after prey animals, tire them out.

- Projecting incisors, slender, parasagittal movement of paws, non-retractable blunt claws.

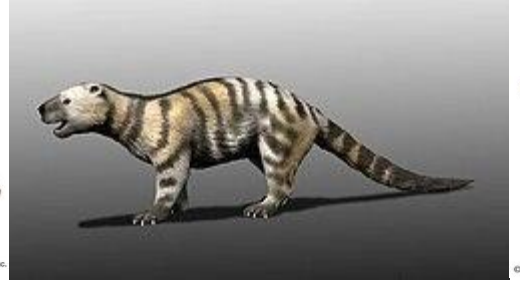


Notable exception – Homotherium (Machairodontidae) with adaptations for cursoriality.

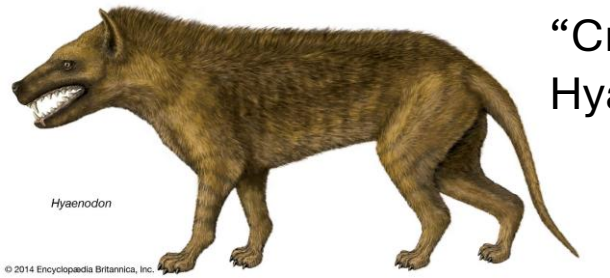


Viverravidae

“Miacidae”

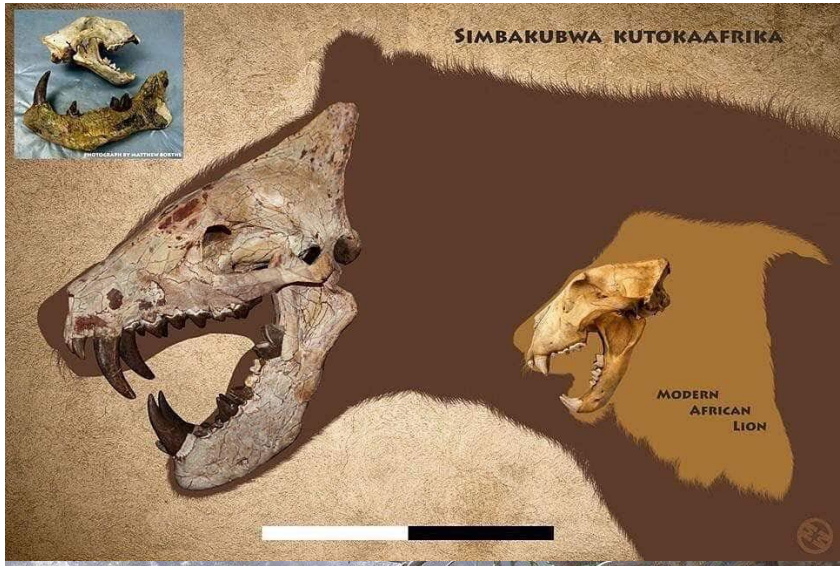


“Creodonta” Oxyaenidae



“Creodonta”
Hyaenodontidae

Simbakubwa kutoafrikaa
(Hyaenodontidae)



STEM CARNIVORA

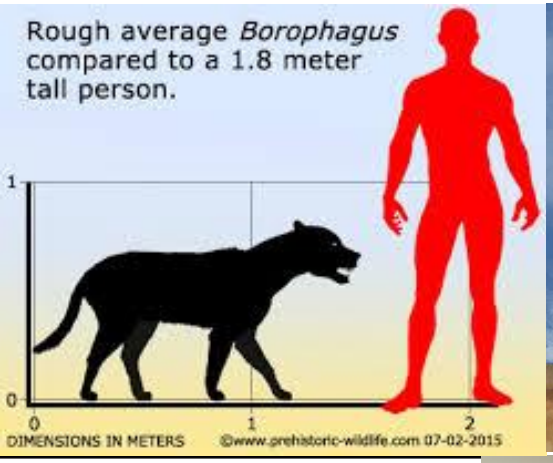
Amphycyonidae “Bear-Dogs”
Stem Caniforma

Clouded Leopard
(Neofelis)
Very similar morphology
to nimravids

Nimravidae (stem
carnivora or stem
feliforma)

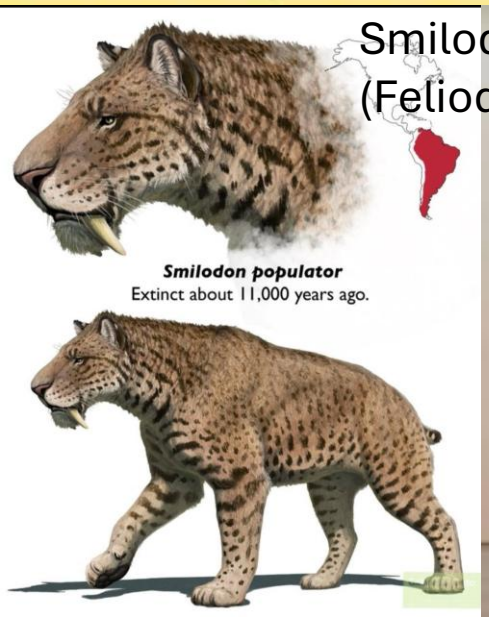
Other (hyper)carnivorous animals of the Cenozoic

“Bone-crushing dogs”



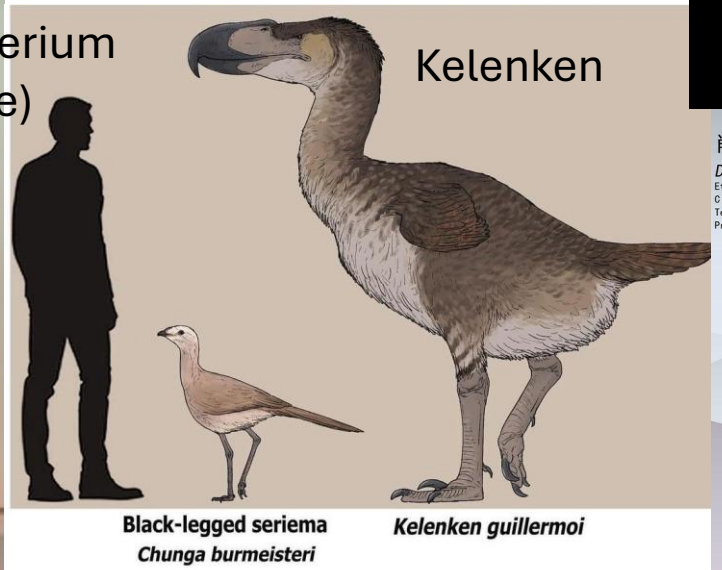
“Megalania” & Quinkana

Andrewsarchus
(Mesonychid)

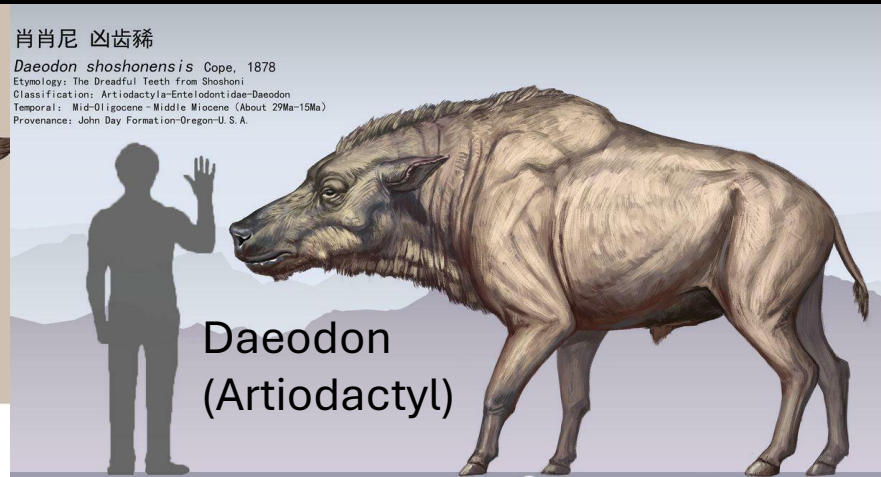


Smilodon
(Feliodea)

Arctotherium
(Ursidae)



Kelenken



Daeodon
(Artiodactyl)